OTC-17 | QUALITY & OPERATIONAL EXCELLENCE

Statistical Process Control & Process Capability

SUGGESTED DURATION: 2 DAYS | IN-COMPANY / LIVE ONLINE

Interpret process data before making improvement decisions. Participants learn when a control chart or capability index is informative and when the underlying assumptions require investigation.

WHO SHOULD ATTEND

QA/QC engineers, production and process engineers, laboratory teams and improvement practitioners familiar with basic arithmetic.

WHAT PARTICIPANTS WILL BE ABLE TO DO

- Distinguish common-cause variation from special-cause signals.
- Select basic variable or attribute charts for an appropriate data structure.
- Interpret Cp/Cpk cautiously and identify prerequisites for meaningful capability analysis.

SUGGESTED PROGRAMME OUTLINE

Day 1 | Data and control charts

Measurement definitions and sampling; rational subgrouping; run charts; introduction to I-MR, X-bar/R and attribute charts; limits versus specifications; signal interpretation and response plans.

Day 2 | Capability and decisions

Stability before capability; specification limits and distribution assumptions; Cp/Cpk interpretation and limitations; measurement variation awareness; stratification and improvement verification; reporting to managers.

OTC-17 | PRACTICAL APPLICATION & DELIVERY

Statistical Process Control & Process Capability

PRACTICAL WORKSHOPS

Control-chart interpretation

Build or interpret an illustrative process chart; identify signals and propose verification rather than automatic process adjustment.

Capability comparison

Compare two fictional processes with different spread and centring; explain why an index alone is insufficient for acceptance.

PARTICIPANT TOOLS AND TAKEAWAYS

Example datasets; chart selection guide; capability review worksheet; out-of-control response-plan template.

HOW LEARNING IS DELIVERED AND ASSESSED

Facilitated explanations, worked examples, case discussions and small-group exercises. A short baseline check establishes starting knowledge; a closing knowledge check and practical review assess learning. Assessment can be adapted for operators, technicians or experienced engineers.

CUSTOMISATION AND CLIENT INPUTS

Anonymised data with units, sampling sequence and specifications. Normal-distribution calculations are introduced with their assumptions; advanced modelling requires separate scope.

The agenda is indicative. Final content, duration, batch size, language and delivery arrangements are agreed after a training-needs discussion. Online delivery uses document reviews, calculations, role-play and case exercises; physical demonstrations require separate site arrangements.

FACILITATOR

Prem Prakash Mishra | Mechanical engineer, advisor and corporate trainer. Brings 39 years of career experience across Kuwait Oil Company, IPCL and NTPC, spanning projects, commissioning, maintenance, reliability, asset integrity and industrial audits. Specialist inputs, where needed, are agreed in the proposal.

COMPLETION AND FOLLOW-UP

An Omtech Corporation participation or completion certificate may be provided under the agreed assessment arrangements. This is company training, not an accredited professional qualification or a guarantee of certification. Post-training support is available with scope and duration agreed in the proposal.

REQUEST A CUSTOMISED TRAINING PROPOSAL

safehands@omtechcorporation.com

+91 9810772284 | +91 9987079111